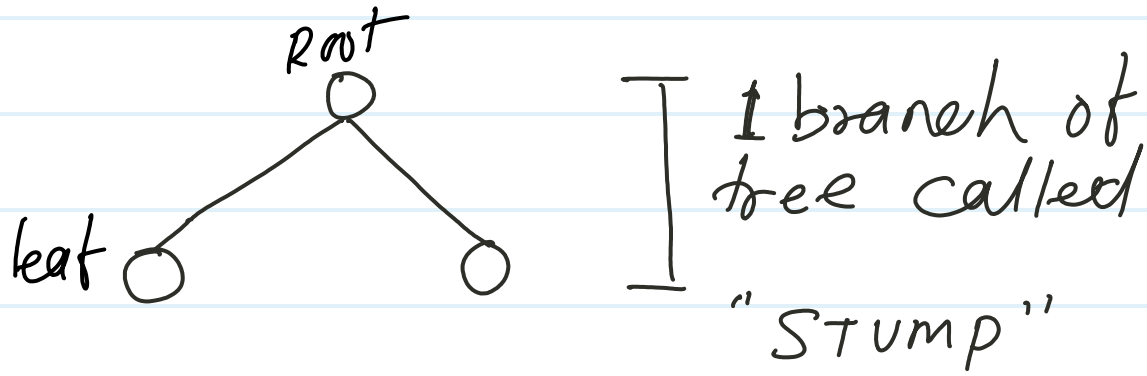
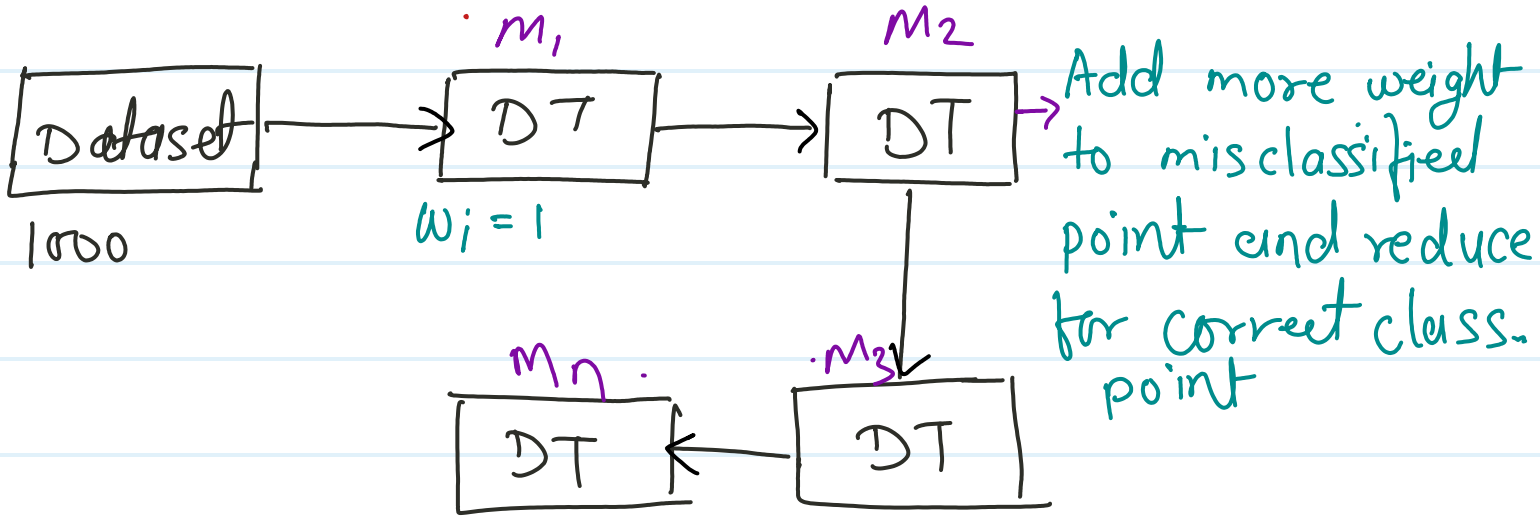
Ada Boost

Same as bagging, Boosting also has homogeneous and heterogeneous model but we use DT only in the ada brosting model.



It is sequencial learning model in this model, every model is known as weaklearner, it pass o/p to the next weaklearner with some weight assigned to it.



Same process occurs for N number of weaklearner based of Combine prediction using a weight majority vote.

AdaBoost commonly used for "classification" problems and very less use for Regression problem.

$$\text{Model} = \alpha_1(m_1) + \alpha_2(m_2) + \alpha_3(m_3) + \cdots + \alpha_n(m_n)$$

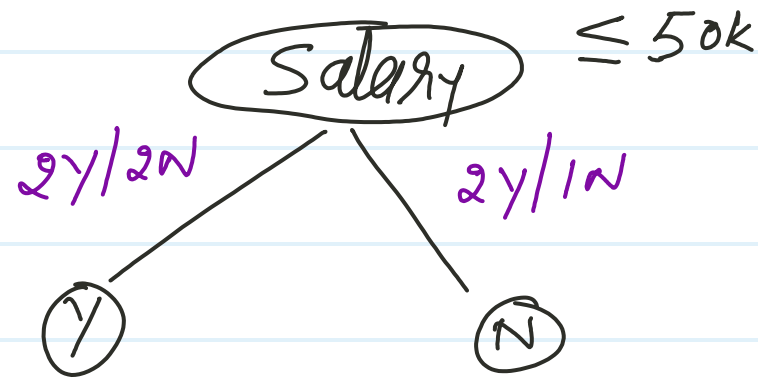
α = weight

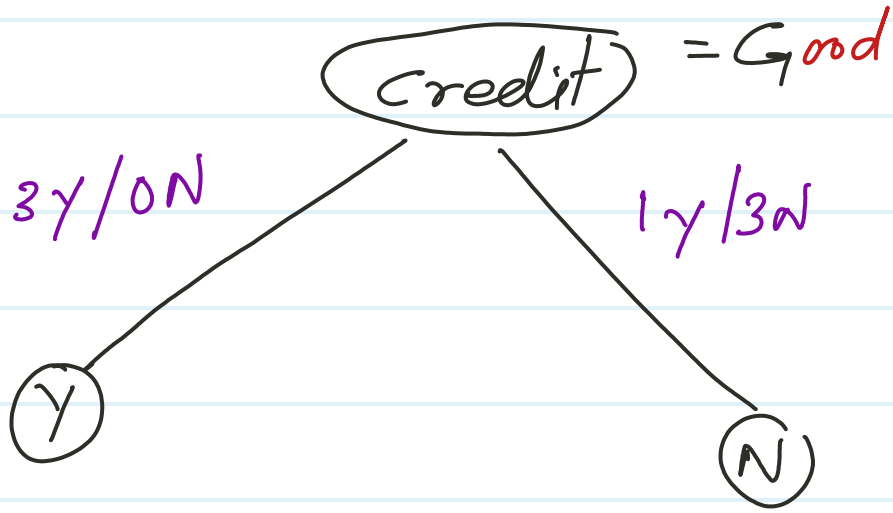
m_1, m_2, m_3, m_n = weaklearner.

Dataset

	Salary (X)	Credit (X)	Approach (Y) loan		weight Ass
1	≤ 50K	Bad	No	$w_i = 1$	1/7
2	≤ 50K	Good	Yes		1/7
3	≤ 50k	Good	Yes		1/7
4	> 50k	Bad	No		1/7
5	> 50k	Good	Yes		1/7
6	> 50k	Neutral	Yes		1/7
7	<= 50k	Neutral	No		1/7

step-1 we create decision tree stump





We use either Entrop or Gini and Find Information Gain

step - 2 After first weaklearner training, how many wrong predicted point identify.

Add more weight to the wrong predicted point (misclassified point)

Reduce weight to the correct classified point.

step - 3

performance of stump $= \frac{1}{2} \ln \left[\frac{1 - T.E}{T.E} \right]$

$$\alpha_1 = \frac{1}{2} \ln \left[\frac{1 - 1/7}{1/7} \right]$$

$$\approx 0.896$$

$\alpha_1 = 0.896$ ✓

step-④ update weight

For correctly classified point

$= \text{weight} \times e^{-\text{Performance of stump}}$

$= \frac{1}{7} \times e^{-(0.896)}$

✓ $= \underline{0.058}$

For incorrect classified point

$= \text{weight} \times e^{\text{Performance of stump}}$

$= 1/7 \times e^{(0.896)}$

$= \underline{0.349}$

For correct data point weight decrease

For incorrect datapoint weight Increase.

Salary (x_1)	Credit (x_2)	Approch (y)	predicted ($\hat{y}$)	updated weight
—	—	Y	Y	0.058
—	—	N	N	0.058
—	—	Y	N	0.349
—	—	N	N	0.058
—	—	Y	Y	0.058
—	—	Y	N	0.349
—	—	Y	Y	0.058

step - 5

Normalization weight computation and assign Bins

Sal	credi.	App.	weight	update weight	Normalization
—	—	—	—	0.058 ÷ 0.988	0.0587
—	—	—	—	0.058 ÷ 0.988	0.0587
—	—	—	—	0.349 ÷ 0.988	0.353
—	—	—	—	0.058 ÷ 0.988	0.0587
—	—	—	—	0.058 ÷ 0.988	0.0587
—	—	—	—	0.349 ÷ 0.988	0.353
—	—	—	—	0.058 ÷ 0.988	0.0587
				0.988	1

Bin Assignment

0 – 0.0587	1
0.0587 – 0.1174	2
0.1174 – 0.4704	3 ✓
0.4704 – 0.5291	4
0.5291 – 0.5878	5
0.5878 – 0.9408	6 ✓
0.9408 – 1	7

Bin size will be big for incorrect data point, so from this method all the incorrect datapoint selected automatically.

And same process will occur for all the weak learner and final weaklearner delivers us correct output.

pros –

① Robust for overfitting condition

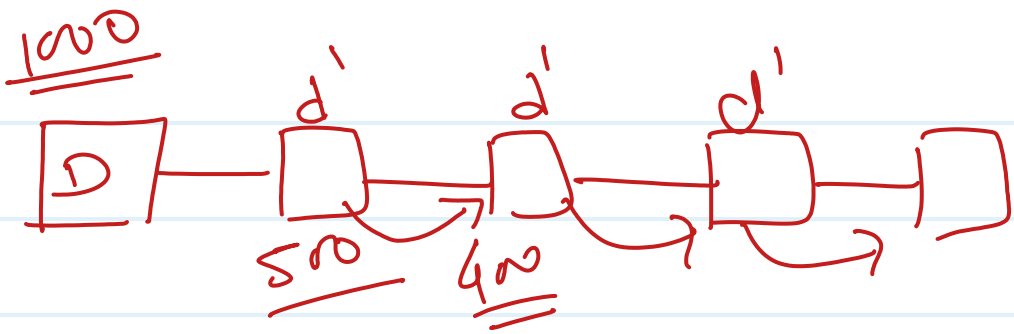
② No feature scaling needed for base tree. (learner)

③ No feature selection needed for base tree (Learner)

Cons –

① Affected by Noisy data

② Complutation requured more time than parallel training



$\boxed{m_1}$ stump performance

1/7 ① correct – weight update

1/7 ② misclass – weight upda

sum

$\frac{\text{correct}}{\text{sum}}$ = Bin size –